

VAIBHAV SIJARIA

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EDUCATION

Vellore Institute of Technology, Vellore

Bachelor of Technology in Computer Science

August 2023 – Present

CGPA 8.68

SKILLS

Programming Languages: Python, Golang, Rust, TypeScript, SQL, Java, C, C++ , Zig

Frameworks & Libraries: NumPy, PyTorch, Flask, Django, Nest.js, Prisma, Drizzle

APIs & Protocols: REST, GraphQL, tRPC, gRPC, WebSockets

Databases: PostgreSQL, MongoDB, Redis, DuckDB

Tools & Technologies: Git, Docker, Linux, GitHub Actions

EXPERIENCE

Xlancr Infotech

September 2024 – February 2025

Software Engineer Intern

Vellore, Tamil Nadu

- Architected scalable, type-safe backend for a high-throughput website and analytics dashboard using **TypeScript**, **NestJS**, and **Drizzle**, capable of successfully handling loads exceeding **5,000 RPM**.
- Reduced API costs and data retrieval latency by **~50%** for a Python backend by engineering a multi-layer caching solution with **Redis** and **FastAPI**; implemented **CI/CD pipeline** via **GitHub Actions** to enforce **70%** test coverage.
- Refactored the codebase to enforce **Separation of Concerns**, leveraging **NestJS's Dependency Injection** patterns to improve scalability, reduce code complexity, and simplify future maintenance.

PROJECTS

Consensus RPC Gateway | *Multi-provider consensus and Merkle proof verification* | [GitHub](#)

March 2026

- Engineered a fault-tolerant **Golang** Ethereum **JSON-RPC** gateway with concurrent multi-provider request dispatch, per-provider timeouts, early quorum exit, and strict-majority consensus-based response routing.
- Validated **eth_getBalance** responses cryptographically using **eth_getProof**, traversing the **Merkle-Patricia Trie** and decoding **RLP** proofs against the block **stateRoot**.
- **Tech Stack:** Golang, Ethereum, RLP

Competitive Coding Platform | *Real-time code submission, evaluation, and leaderboard* | [GitHub](#)

October 2024

- Achieved **100% uptime** while serving **600+ users** in a live competition. Implemented **Redis** caching for the duplicate submissions and contest questions, significantly reducing direct database queries and code judge load.
- Architected a secure and scalable code evaluation pipeline by integrating **Judge0** for sandboxed execution and **Asyncq** for asynchronous job queuing, ensuring fast, non-blocking submission processing.
- **Tech Stack:** Golang, Docker, PostgreSQL, Redis, Asyncq, SQLC, Judge0, Gin framework

Distributed MapReduce Framework | *Language-agnostic MapReduce engine* | [GitHub](#)

May 2026

- Designed a **language-agnostic** execution model that compiles user-supplied Map/Reduce source and runs it via a **stdin/stdout** key-value contract, with in-memory partition buffers that spill to disk past a threshold.
- Built crash-safe recovery via periodic **JSON checkpointing** of job/task state to S3, ephemeral worker re-registration on heartbeat loss, and **atomic output promotion** via S3 CopyObject/DeleteObject.
- **Tech Stack:** Golang, gRPC, Protocol Buffers, AWS S3, Docker, HTTP

LEADERSHIP & EXTRACURRICULAR

Tech Coordinator, Gravitas 2025

September 2025

- Engineered high-throughput payment workflow using **Go**, **PostgreSQL**, and a **Redis**-based queue system. Successfully handled **3 million+ visits** and processed over **50 lakhs** in transactions with **zero downtime** during peak load.
- Eliminated booking related **race conditions** via a 10-minute seat-hold system based on **Redis Sorted Sets**.
- Resolved a critical transaction leak on all code paths, preventing **connection pool exhaustion**; also shipped new accommodation/check-in workflows and built admin dashboards, which reduced support tickets by **95%**.

Event Sub-Coordinator, CookOff 9.0 - Gravitas 2024

September 2024

- Led a 5-member backend team in developing the submission portal for **Cook Off 9.0**, which processed **15,000+ evaluations** with **zero downtime**, a peak throughput of **200+ submissions/min**, and an average verdict time of **<10s**.

1st Runner up, ForkThis - Gravitas 2023

September 2023

- Contributed to open-source projects by resolving bugs and implementing features across multiple repositories. Secured **2nd place out of 300+ participants** with **25 merged PRs**, utilized **Python**, **JavaScript**, **Selenium**, and **Node.js**.